Bihar Engineering University, Patna

B.Tech 6th Semester Examination, 2025

Course: B.Tech | **Time: 03 Hours**
Code: 101604 | **Subject: Engineering Economics, Estimation & Costing** | **Full Marks: 70**

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are **NINE** questions in this paper.
(iii) Attempt **FIVE** questions in all.
(iv) Question No. **1** is compulsory.

Q.1 Choose the correct option / answer the following (Any seven question only): [2 x 7 = 14]

(a) Which of the following is NOT a type of elasticity of demand?
(i) Price elasticity
(ii) Cross elasticity
(iii) Income elasticity
(iv) Capital elasticity

(b) GDP at market price includes
(i) Net factor income from abroad
(ii) Direct taxes only
(iii) Indirect taxes minus subsidies
(iv) Depreciation only

(c) Break-even point is achieved when
(i) Total cost = Total revenue
(ii) Profit = Fixed cost
(iii) Fixed cost = Variable cost
(iv) Variable cost = Total cost

(d) The unorganized sector in India is characterized by
(i) Job security and benefits
(ii) Lack of job security and irregular income
(iii) Regular wages
(iv) Large scale industries

(e) The estimated quantity of cement in 1 m^3 of concrete having nominal mix 1:2:4 by thumb rule is
(i) 2 bags
(ii) 6 bags
(iii) 4 bags
(iv) 8 bags

(f) Detailed specification includes
(i) General guidelines only
(ii) Only quantity details
(iii) Material, workmanship, and method of execution
(iv) None of the above

(g) The purpose of rate analysis is
(i) To fix the selling price of a product
(ii) To measure productivity only
(iii) To calculate total project cost per unit of work
(iv) To prepare bar chart

(h) A tender is
(i) A payment certificate
(ii) A project report
(iii) An offer to execute work at specified rates
(iv) A technical drawing

(i) The primary purpose of inviting tenders is to ensure
(i) Profit to contractor
(ii) Transparency and competition
(iii) Minimum time for project execution
(iv) Reduction of material cost

(j) The most used method for dispute settlement in construction projects is
(i) Meditation
(ii) Arbitration
(iii) Court litigation
(iv) Public voting

© Apna BEU

Apna BEU: Syllabus, PYQs & AI
All-in-One app for Syllabus, PYQs, Result, Attendance and more.
GET IT ON Google Play

Q.2 (a) Explain the concept of Aggregate Demand (AD) and Aggregate Supply (AS) in the context of an open economy. [7]

(b) What is Contract? Explain different types of Contracts [7]

Q.3 (a) Explain how managerial economics helps in managerial decision-making in an organization. [7]

(b) Explain how break-even point helps managers in taking production decisions. [7]

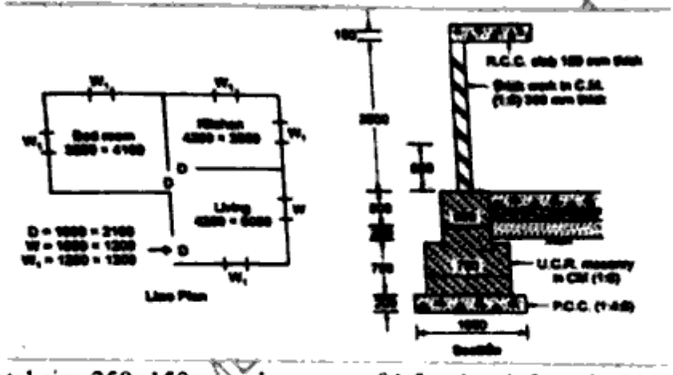
Q.4 (a) Explain moving average forecasting with a simple illustration. [7]

(b) Explain why balance sheet is called a "snapshot" of the financial position of a business. [7]

Q.5 (a) Discuss the changing contribution of the primary, secondary, and tertiary sectors to India's GDP after reforms. [7]

(b) Describe the role of direct benefit transfer (DBT) in improving social sector outcomes. [7]

Q.6 (a) Estimate the quantities using Centre line method [7]
(1) Earthwork in excavation in foundation
(2) P.C.C in foundation
(3) Ist class Brickwork in cement mortar 1:6 in superstructure.



(b) A RCC lintel size 250×150 mm clear span of 1.5 m is reinforced with 4 bars of 10 mm dia. at bottom and 3 bars of 8 mm at top. The stirrups of 6 mm dia. are provided 150 mm c/c. Bearing of lintel is 150 mm. Calculate the total quantity of steel reinforcement. [7]

Q.7 (a) Estimate the quantities of materials required for a RCC roof slab 12 cm thick with1% reinforcement for a room having internal dimension 4.5 m×3.5 m. Estimate the cost of construction of the roof slab with suitable rates of materials and taking labour rate of laying including centering and shuttering etc. as ₹80.00 per cum of RCC work. [7]

(b) Prepare rate analysis of for plastering 12 mm thick in cement mortar 1:3. Assume suitable data if needed. [7]

Q.8 (a) Calculate The quantity of earthwork for 200 m length for a portion of a road in a uniform ground height of bank at the two ends being 1m and 1.6 m. The formation width is 10 m and side slope 2:1 (Horizontal: Vertical). Assume that there is no transverse slope. [7]

(b) Write the rules for deduction in painting work for doors and windows as per IS 1200. Also discuss the important points which should be considered while taking measurements. [7]

Q.9 Writes Short notes on any four of the following: **[3.5 x 4=14]**

(a) Termination of Contracts
(b) Direct & Indirect Overheads
(c) Importance of inviting tenders
(d) Detailed specification for industrial structures
(e) Factors affecting task work
(f) Penalty and Liquidated charges in contracts

HIGHLY CONFIDENTIAL DOCUMENT
© Apna BEU
1-2025 13:37:06

Apna BEU: Syllabus, PYQs & AI
All-in-One app for Syllabus, PYQs, Result, Attendance and more.
GET IT ON Google Play